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POLAR CODING TECHNIQUES FOR HIGHER-ORDER MODULATION

Abstract

Methods, systems, and devices for wireless communications are described. In some examples, a transmitting device, such as a user equipment (UE) or a network entity, may encode an information bit vector into multiple code blocks using a polar code partitioned into multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each code block of the multiple code blocks may be encoded using a respective sub-block of the multiple sub-blocks. Additionally, the transmitting device may interleave respective bits of each code block of the multiple code blocks using a respective interleaver of multiple interleavers. As such, the transmitting device may transmit the respective interleaved bits of each code block of the multiple code blocks over a respective bit channel of multiple bit channels, where each bit channel of the multiple bit channels is associated with a respective reliability.

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Background/Summary

CROSS REFERENCE [0001] The present Application for Patent is a division of U.S. patent application Ser. No. 18/359,235 by IVANOV et al., entitled “POLAR CODING TECHNIQUES FOR HIGHER-ORDER MODULATION SCHEMES,” filed Jul. 26, 2023, assigned to the assignee hereof, and is expressly incorporated by reference in its entirety herein.

TECHNICAL FIELD

[0002] The following relates to wireless communication and, more specifically polar coding techniques for higher-order modulation schemes.

DESCRIPTION OF THE RELATED TECHNOLOGY

[0003] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (such as, time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems, which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).

[0004] Some wireless communications systems may support the use of polar coding for efficient and relatively low complexity decoding of wireless communications. Polar coding involves assigning information bits to different bit channels and encoding the bits such that some bit channels are associated with increased reliability and other bit channels are associated with decreased reliability (such that respective bit channels may be “polarized” in terms of reliability). An encoder of a transmitting device may implement a polar code using a matrix associated with the polar code that polarizes copies of a channel into subchannels (which may be referred to as bit channels), which are either relatively noisy (corresponding to decreased reliability) or relatively noiseless (corresponding to increased reliability). Information bits may be mapped to the noiseless bit channels and frozen bits may be mapped to the noisy bit channels, and the transmitting device may transmit a codeword in accordance with the mapping to the respective bit channels. In some examples, rate matching may be performed on a codeword, which may include puncturing or shortening some quantity of coded bits of the codeword.

SUMMARY

[0005] The systems, methods, and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

[0006] One innovative aspect of the subject matter described in this disclosure can be implemented in a method for wireless communication. The method includes encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a

respective sub-block of the set of multiple sub-blocks, interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers, and transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0007] Another innovative aspect of the subject matter described in this disclosure can be implemented in a wireless communication device. The wireless communication device may include a processing system that includes processor circuitry and memory circuitry that stores code. The processing system may be configured to cause the wireless communication device to encode an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks, interleave respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers, and transmit the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0008] Another innovative aspect of the subject matter described in this disclosure can be implemented in another wireless communication device. The wireless communication device may include means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks, means for interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers, and means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0009] Another innovative aspect of the subject matter described in this disclosure can be implemented in a non-transitory computer-readable medium storing code for wireless communication is described. The code may include instructions executable by one or more processors to encode an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks, interleave respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers, and transmit the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0010] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, encoding the information bit vector may include operations, features, means, or instructions for mapping bits of the information bit vector to respective bit locations associated with the set of multiple sub-blocks, where the bits of each code block may be interleaved in accordance with the respective bit locations.

[0011] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, encoding the information bit vector may include operations, features, means, or instructions for allocating a respective quantity of bits of the information bit vector to each sub-block of the set of multiple sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector and determining a bit

position for each bit of the respective quantity of bits allocated to each sub-block of the set of multiple sub-blocks, where each code block of the set of multiple code blocks includes the respective quantity of bits of the information bit vector in accordance with the bit positions.

[0012] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, allocating the quantities of bits of the information bit vector to the set of multiple sub-blocks may include operations, features, means, or instructions for determining single-bit reliabilities of respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks, selecting respective subsets of the respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with the single-bit reliabilities, and dividing respective sets of bits into partitions allocated to the respective subsets of the respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with the single-bit reliabilities.

[0013] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, determining the bit position for each bit of the respective quantity of bits may include operations, features, means, or instructions for determining the bit position for each bit of the partitions in each bit channel sub-block in accordance with the single-bit reliabilities.

[0014] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, allocating the respective quantity of bits of the information bit vector to each sub-block of the set of multiple sub-blocks may include operations, features, means, or instructions for dividing respective sets of bits into partitions allocated to bit channel sub-blocks for each sub-block of the set of multiple sub-blocks, where the respective sets of bits may be divided in accordance with a mutual information transfer function, and where the respective quantity of bits may be allocated to each sub-block of the set of multiple sub-blocks in accordance with the mutual information transfer function.

[0015] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, the respective sets of bits may be divided into partitions recursively, the recursive dividing may be stopped based on the bit channel sub-blocks being associated with a threshold length, and determining the bit position for each bit of the respective quantity of bits may be based on the bit channel sub-blocks being associated with the threshold length.

[0016] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, each code block may be associated with a respective quantity of encoded bits and the method, apparatuses, and non-transitory computer-readable medium may include further operations, features, means, or instructions for performing rate matching for the respective quantity of the encoded bits associated with each code block in accordance with the same block length of each sub-block.

[0017] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, performing the rate matching for the respective quantity of the encoded bits may include operations, features, means, or instructions for shortening a subset of the respective quantity of the encoded bits associated with each code block, the subset including a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with each code block.

[0018] In some examples of the method, wireless communication devices, and non-transitory computer-readable medium described herein, performing the rate matching for the respective quantity of the encoded bits may include operations, features, means, or instructions for puncturing a subset of the respective quantity of the encoded bits associated with each code block, the subset including a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with each code block.

[0019] Details of one or more implementations of the subject matter described in this disclosure are

set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 shows a pictorial diagram of an example wireless communications system.

[0021] FIG. 2 shows an example of a wireless communications system that supports polar coding techniques for higher-order modulation schemes.

[0022] FIGS. 3 and 4 each show examples of rate matching schemes that support polar coding techniques for higher-order modulation schemes.

[0023] FIG. 5 shows an example of a process flow that supports polar coding techniques for higher-order modulation schemes.

[0024] FIGS. 6 and 7 show block diagrams of example wireless communication devices that support polar coding techniques for higher-order modulation schemes.

[0025] FIG. 8 shows a block diagram of an example communications manager that supports polar coding techniques for higher-order modulation schemes.

[0026] FIGS. 9 and 10 show diagrams of an example system including a wireless communication device that supports polar coding techniques for higher-order modulation schemes.

[0027] FIGS. 11 and 12 show flowcharts illustrating methods that support polar coding techniques for higher-order modulation schemes.

[0028] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0029] The following description is directed to some implementations for the purposes of describing the innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways. The described implementations may be implemented in any device, system, or network that is capable of transmitting and receiving radio frequency (RF) signals according to any of the Institute of Electrical and Electronics Engineers (IEEE) 16.11 standards, or any of the IEEE 802.11 standards, the Bluetooth® standard, code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), Global System for Mobile communications (GSM), GSM/General Packet Radio Service (GPRS), Enhanced Data GSM Environment (EDGE), Terrestrial Trunked Radio (TETRA), Wideband-CDMA (W-CDMA), Evolution Data Optimized (EV-DO), 1xEV-DO, EV-DO Rev A, EV-DO Rev B, High Speed Packet Access (HSPA), High Speed Downlink Packet Access (HSDPA), High Speed Uplink Packet Access (HSUPA), Evolved High Speed Packet Access (HSPA+), Long Term Evolution (LTE), AMPS, or other known signals that are used to communicate within a wireless, cellular or internet of things (IOT) network, such as a system utilizing third generation (3G), fourth generation (4G), fifth generation (5G), or sixth generation (6G), or further implementations thereof, technology.

[0030] Some wireless communications systems may support the use of polar codes for wireless communications. Polar coding involves assigning information bits to different bit channels and encoding the bits such that some bit channels are associated with relatively increased reliability and other bit channels are associated with relatively decreased reliability (such that respective bit channels may be polarized). As an example, an encoder of a transmitting wireless communication device (such as a user equipment (UE) or a network entity) may implement a polar code using a matrix that polarizes copies of a channel into subchannels (which may be referred to as bit channels) that are either relatively noisy (corresponding to decreased reliability) or relatively noiseless (corresponding to increased reliability). Information bits may be mapped to the relatively

noiseless bit channels, and frozen bits (for example, bits that may be known by both the transmitting wireless communication device and a receiving wireless communication device) may be mapped to the relatively noisy bit channels. The transmitting wireless communication device may transmit a codeword in accordance with the mapping to the respective bit channels. In some cases, a block length of the polar code may be associated with subchannel reliabilities that may be optimal for some modulation schemes (such as quadrature phase shift keying (QPSK) modulation schemes) but not others. For example, bits of some relatively higher-order modulation scheme symbols (such as quadrature amplitude modulation (QAM) symbols) may be associated with different reliabilities, and some polar coding techniques may not consider these bit reliabilities when information bits are encoded. For instance, some polar coding techniques may average bit reliabilities, which may ignore the reliabilities of the bits in, for example, QAM symbols.

[0031] Various aspects generally relate to polar coding techniques for higher-order modulation schemes, and more specifically, to techniques that use different bit reliability levels for symbols of higher-order modulation schemes (such as QAM and others) in which a polar code is partitioned (for example, divided) into multiple sub-blocks that each have a similar structure. In some examples, a transmitting wireless communication device may apply a polar transform (for example, using a polar transformation matrix) to a polar code of block length N that partitions the polar code into the multiple sub-blocks that have a same block length (for example, a block length of $N/4$ for a 256 QAM scheme). Partitioning the polar code into the respective sub-block may be in accordance with an order of a modulation scheme, and the respective sub-blocks may each correspond to a different bit channel having a corresponding reliability. The transmitting device may assign information bits of an information bit vector to bit channels using the multiple sub-blocks, resulting in respective code blocks. The respective code blocks may correspond to different bit locations (such as a most significant bit, a least significant bit, and others), and the transmitting device may independently interleave bits of each code block with a respective bit-interleaver to obtain a codeword for transmission over the bit channels. In some aspects, rate matching techniques, such as shortening and puncturing, may be used for the codewords generated using the multiple sub-blocks of the partitioned polar code.

[0032] Particular aspects of the subject matter described in the disclosure may be implemented to realize one or more of the following advantages. The techniques employed by the described wireless communication devices, such as the transmitting wireless communication device and the receiving wireless communication device, may provide benefits and enhancements to the operation of the wireless communication devices, including efficient encoding and decoding, providing reduced latency and power consumption. For example, operations performed by the described communication devices may provide improvements to polar coding procedures by utilizing different bit reliability levels of QAM symbols, and polar coding performed on such symbols may take advantage of such reliability levels to enhance a code design (for example, by providing improved polarization due to bit grouping). In such examples, the operations performed by the described wireless communication devices using different bit reliability levels of QAM symbols may improve the reliability of transmissions encoded using polar codes.

[0033] FIG. 1 shows a pictorial diagram of an example wireless communications system **100**. The wireless communications system **100** may include one or more network entities **105**, one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0034] The network entities **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network

element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities **105** and UEs **115** may wirelessly communicate via one or more communication links **125** (such as a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (such as a geographic coverage area) over which the UEs **115** and the network entity **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

[0035] The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. 1. The UEs **115** described herein may be capable of supporting communications with various types of devices, such as other UEs **115** or network entities **105**, as shown in FIG. 1.

[0036] As described herein, a node of the wireless communications system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (such as any network entity described herein), a UE **115** (such as any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, among other examples, may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, among other examples, being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

[0037] In some examples, network entities **105** may communicate with the core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via one or more backhaul communication links **120** (such as in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities **105** may communicate with one another via a backhaul communication link **120** (such as in accordance with an X2, Xn, or other interface protocol) either directly (such as directly between network entities **105**) or indirectly (such as via a core network **130**). In some examples, network entities **105** may communicate with one another via a midhaul communication link **162** (such as in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (such as in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication links **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (such as an electrical link, an optical fiber link), one or more wireless links (such as a radio link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** via a communication link **155**.

[0038] One or more of the network entities **105** described herein may include or may be referred to as a base station (BS) **140** (such as a base transceiver station, a radio BS, an NR BS, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity **105** (such as a BS **140**) may be implemented in an aggregated (such as monolithic, standalone) BS

architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within a single network entity **105** (such as a single RAN node, such as a BS **140**). [0039] In some examples, a network entity **105** may be implemented in a disaggregated architecture (such as a disaggregated BS architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among two or more network entities **105**, such as an integrated access backhaul (IAB) network, an open RAN (O-RAN) (such as a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (such as a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU) **160**, a distributed unit (DU) **165**, a radio unit (RU) **170**, a RAN Intelligent Controller (RIC) **175** (such as a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) **180** system, or any combination thereof. An RU **170** also may be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (such as separate physical locations). In some examples, one or more network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (such as a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0040] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending on which functions (such as network layer functions, protocol layer functions, baseband functions, RF functions, and any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (such as layer 3 (L3), layer 2 (L2)) functionality and signaling (such as Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** may be connected to one or more DUs **165** or RUs **170**, and the one or more DUs **165** or RUs **170** may host lower protocol layers, such as layer 1 (L1) (such as physical (PHY) layer) or L2 (such as radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (such as via one or more RUs **170**). In some examples, a functional split between a CU **160** and a DU **165**, or between a DU **165** and an RU **170** may be within a protocol layer (such as some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to one or more DUs **165** via a midhaul communication link **162** (such as F1, F1-c, F1-u), and a DU **165** may be connected to one or more RUs **170** via a fronthaul communication link **168** (such as open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (such as a channel) between layers of a protocol stack supported by respective network entities **105** that are in communication via such communication links.

[0041] In wireless communications systems (such as wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (such as to a core network **130**). In some examples, in an IAB network, one or more network

entities **105** (such as IAB nodes **104**) may be partially controlled by each other. One or more IAB nodes **104** may be referred to as a donor entity or an IAB donor. One or more DUs **165** or one or more RUs **170** may be partially controlled by one or more CUs **160** associated with a donor network entity **105** (such as a donor BS **140**). The one or more donor network entities **105** (such as IAB donors) may be in communication with one or more additional network entities **105** (such as IAB nodes **104**) via supported access and backhaul links (such as backhaul communication links **120**). IAB nodes **104** may include an IAB mobile termination (IAB-MT) controlled (such as scheduled) by DUs **165** of a coupled IAB donor. An IAB-MT may include an independent set of antennas for relay of communications with UEs **115**, or may share the same antennas (such as of an RU **170**) of an IAB node **104** used for access via the DU **165** of the IAB node **104** (such as referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB nodes **104** may include DUs **165** that support communication links with additional entities (such as IAB nodes **104**, UEs **115**) within the relay chain or configuration of the access network (such as downstream). In such examples, one or more components of the disaggregated RAN architecture (such as one or more IAB nodes **104** or components of IAB nodes **104**) may be configured to operate according to the techniques described herein.

[0042] In the examples of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support polar coding techniques for higher-order modulation schemes as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (such as a BS **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (such as IAB nodes **104**, DUs **165**, CUs **160**, RUs **170**, RIC **175**, SMO **180**).

[0043] A UE **115** may include or may be referred to as a mobile device, a wireless device, a wireless communication device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” also may be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** also may include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.

[0044] The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes function as relays as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay BSs, among other examples, as shown in FIG. 1.

[0045] The UEs **115** and the network entities **105** may wirelessly communicate with one another via one or more communication links **125** (such as an access link) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a RF spectrum band (such as a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (such as LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (such as synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division

duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (such as entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (such as a BS **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (such as directly or via one or more other network entities **105**).

[0046] The communication links **125** shown in the wireless communications system **100** may include downlink transmissions (such as forward link transmissions) from a network entity **105** to a UE **115**, uplink transmissions (such as return link transmissions) from a UE **115** to a network entity **105**, or both, among other configurations of transmissions. Carriers may carry downlink or uplink communications (such as in an FDD mode) or may be configured to carry downlink and uplink communications (such as in a TDD mode).

[0047] A carrier may be associated with a particular bandwidth of the RF spectrum and, in some examples, the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a set of bandwidths for carriers of a particular radio access technology (such as 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (such as the network entities **105**, the UEs **115**, or both) may have hardware configurations that support communications using a particular carrier bandwidth or may be configurable to support communications using one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include network entities **105** or UEs **115** that support concurrent communications using carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating using portions (such as a sub-band, a BWP) or all of a carrier bandwidth.

[0048] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (such as using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (such as a duration of one modulation symbol) and one subcarrier, for which the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (such as the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (such as in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (such as a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

[0049] Modulation is the process of representing a digital signal by modifying the properties of a periodic waveform (such as, frequency, amplitude, and phase). Demodulation takes a modified waveform and generates a digital signal. A modulated waveform may be divided into time units known as symbols. Each symbol may be modulated separately. In a wireless communication system (such as wireless communications system **100**) that uses relatively frequency subcarriers to transmit distinct symbols, the modulation may be accomplished by varying the phase and amplitude of each symbol. For example, a binary phase shift keying (BPSK) modulation scheme conveys information by alternating between waveforms that are transmitted with no phase offset or with a 180° offset (that is, each symbol conveys a single bit of information). In a QAM scheme, two carrier signals (known as the in-phase component, I, and the quadrature component, Q) may be transmitted with a phase offset of 90°, and each signal may be transmitted with specific amplitude selected from a finite set. The number of amplitude bins may determine the number of bits that are conveyed by each symbol. For example, in a 16-QAM scheme, each carrier signal may have one of four amplitudes (such as, -3, -1, 1, 3), which results in 16 possible combinations (that is, 4 bits).

The various possible combinations may be represented in a graph known as a constellation map, where the amplitude of the I component is represented on the horizontal axis and the Q component is represented on the vertical axis.

[0050] One or more numerologies for a carrier may be supported, and a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and communications for the UE **115** may be restricted to one or more active BWPs.

[0051] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, in some examples, refer to a sampling period of $T_{\text{sub.s}} = 1/(\Delta f_{\text{sub.max}} \cdot N_{\text{sub.f}})$ seconds, for which $\Delta f_{\text{sub.max}}$ may represent a supported subcarrier spacing, and $N_{\text{sub.f}}$ may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (such as 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (such as ranging from 0 to 1023).

[0052] Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (such as in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (such as depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (such as $N_{\text{sub.f}}$) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0053] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (such as in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (such as a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (such as in bursts of shortened TTIs (STTIs)).

[0054] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (such as a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (such as CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (such as control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**.

[0055] A network entity **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a

network entity **105** (such as using a carrier) and may be associated with an identifier for distinguishing neighboring cells (such as a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell also may refer to a coverage area **110** or a portion of a coverage area **110** (such as a sector) over which the logical communication entity operates. Such cells may range from smaller areas (such as a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the network entity **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with coverage areas **110**, among other examples.

[0056] A macro cell generally covers a relatively large geographic area (such as several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered network entity **105** (such as a lower-powered BS **140**), as compared with a macro cell, and a small cell may operate using the same or different (such as licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (such as the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A network entity **105** may support one or multiple cells and also may support communications via the one or more cells using one or multiple component carriers.

[0057] In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (such as MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

[0058] In some examples, a network entity **105** (such as a BS **140**, an RU **170**) may be movable and provide communication coverage for a moving coverage area **110**. In some examples, different coverage areas **110** associated with different technologies may overlap, but the different coverage areas **110** may be supported by the same network entity **105**. In some other examples, the overlapping coverage areas **110** associated with different technologies may be supported by different network entities **105**. The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** provide coverage for various coverage areas **110** using the same or different radio access technologies.

[0059] Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (such as via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a network entity **105** (such as a BS **140**) without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that uses the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

[0060] Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (such as a mode that supports one-way communication via transmission or reception, but not transmission and reception concurrently). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for the UEs **115** include entering a power saving deep sleep mode when not engaging in active communications, operating using a limited bandwidth (such as according to narrowband communications), or a combination of these techniques. For example,

some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a defined portion or range (such as set of subcarriers or resource blocks (RBs)) within a carrier, within a guard-band of a carrier, or outside of a carrier.

[0061] The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0062] In some examples, a UE **115** may be configured to support communicating directly with other UEs **115** via a device-to-device (D2D) communication link **135** (such as in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some examples, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (such as a BS **140**, an RU **170**), which may support aspects of such D2D communications being configured by (such as scheduled by) the network entity **105**. In some examples, one or more UEs **115** of such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some examples, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in which each UE **115** transmits to each of the other UEs **115** in the group. In some examples, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs **115** without an involvement of a network entity **105**.

[0063] In some systems, a D2D communication link **135** may be an example of a communication channel, such as a sidelink communication channel, between vehicles (such as UEs **115**). In some examples, vehicles may communicate using vehicle-to-everything (V2X) communications, vehicle-to-vehicle (V2V) communications, or some combination of these. A vehicle may signal information related to traffic conditions, signal scheduling, weather, safety, emergencies, or any other information relevant to a V2X system. In some examples, vehicles in a V2X system may communicate with roadside infrastructure, such as roadside units, or with the network via one or more network nodes (such as network entities **105**, BSs **140**, RUs **170**) using vehicle-to-network (V2N) communications, or with both.

[0064] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (such as a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (such as a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (such as BSs **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0065] The wireless communications system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region

from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. Communication using UHF waves may be associated with smaller antennas and shorter ranges (such as less than 100 kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0066] The wireless communications system **100** also may operate using a super high frequency (SHF) region, which may be in the range of 3 GHz to 30 GHz, also known as the centimeter band, or using an extremely high frequency (EHF) region of the spectrum (such as from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, the wireless communications system **100** may support millimeter wave (mmW) communications between the UEs **115** and the network entities **105** (such as BSs **140**, RUs **170**), and EHF antennas of the respective devices may be smaller and more closely spaced than UHF antennas. In some examples, such techniques may facilitate using antenna arrays within a device. The propagation of EHF transmissions, however, may be subject to even greater attenuation and shorter range than SHF or UHF transmissions. The techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.

[0067] The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be in accordance with a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (such as LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0068] A network entity **105** (such as a BS **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more BS antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0069] The network entities **105** or the UEs **115** may use MIMO communications to exploit multipath signal propagation and increase spectral efficiency by transmitting or receiving multiple signals via different spatial layers. Such techniques may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry information associated with

the same data stream (such as the same codeword) or different data streams (such as different codewords). Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO), for which multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO), for which multiple spatial layers are transmitted to multiple devices.

[0070] Beamforming, which also may be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (such as a network entity **105**, a UE **115**) to shape or steer an antenna beam (such as a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (such as with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0071] A network entity **105** or a UE **115** may use beam sweeping techniques as part of beamforming operations. For example, a network entity **105** (such as a BS **140**, an RU **170**) may use multiple antennas or antenna arrays (such as antenna panels) to conduct beamforming operations for directional communications with a UE **115**. Some signals (such as synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a network entity **105** multiple times along different directions. For example, the network entity **105** may transmit a signal according to different beamforming weight sets associated with different directions of transmission. Transmissions along different beam directions may be used to identify (such as by a transmitting device, such as a network entity **105**, or by a receiving device, such as a UE **115**) a beam direction for later transmission or reception by the network entity **105**.

[0072] Some signals, such as data signals associated with a particular receiving device, may be transmitted by transmitting device (such as a transmitting network entity **105**, a transmitting UE **115**) along a single beam direction (such as a direction associated with the receiving device, such as a receiving network entity **105** or a receiving UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined according to a signal that was transmitted along one or more beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the network entity **105** along different directions and may report to the network entity **105** an indication of the signal that the UE **115** received with a highest signal quality or an otherwise acceptable signal quality.

[0073] In some examples, transmissions by a device (such as by a network entity **105** or a UE **115**) may be performed using multiple beam directions, and the device may use a combination of digital precoding or beamforming to generate a combined beam for transmission (such as from a network entity **105** to a UE **115**). The UE **115** may report feedback that indicates precoding weights for one or more beam directions, and the feedback may correspond to a configured set of beams across a system bandwidth or one or more sub-bands. The network entity **105** may transmit a reference signal (such as a cell-specific reference signal (CRS), a channel state information reference signal (CSI-RS)), which may be precoded or unprecoded. The UE **115** may provide feedback for beam selection, which may be a precoding matrix indicator (PMI) or codebook-based feedback (such as a multi-panel type codebook, a linear combination type codebook, a port selection type codebook). Although these techniques are described with reference to signals transmitted along one or more directions by a network entity **105** (such as a BS **140**, an RU **170**), a UE **115** may employ similar techniques for transmitting signals multiple times along different directions (such as for identifying

a beam direction for subsequent transmission or reception by the UE **115**) or for transmitting a signal along a single direction (such as for transmitting data to a receiving device).

[0074] A receiving device (such as a UE **115**) may perform reception operations in accordance with multiple receive configurations (such as directional listening) when receiving various signals from a transmitting device (such as a network entity **105**), such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may perform reception in accordance with multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets (such as different directional listening weight sets) applied to signals received at multiple antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at multiple antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive configurations or receive directions. In some examples, a receiving device may use a single receive configuration to receive along a single beam direction (such as when receiving a data signal). The single receive configuration may be aligned along a beam direction determined in accordance with listening according to different receive configuration directions (such as a beam direction determined to have a highest signal strength, highest signal-to-noise ratio (SNR), or otherwise acceptable signal quality in accordance with listening according to multiple beam directions).

[0075] The UEs **115** and the network entities **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly via a communication link (such as a communication link **125**, a D2D communication link **135**). HARQ may include a combination of error detection (such as using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (such as automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (such as low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, for which the device may provide HARQ feedback in a specific slot for data received via a previous symbol in the slot. In some other examples, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

[0076] The wireless communications system **100** may support techniques for partitioning a polar code into multiple sub-blocks (such as polar code sub-blocks), such that a transmitting wireless communication device may utilize different reliabilities of respective bits of a modulation symbol when encoding an information bit vector into multiple code blocks. For example, a transmitting wireless communication device, such as a UE **115** or a network entity **105**, may apply a polar transform (such as a transformation matrix) to a polar code (for example, a polar code of block length N) to partition the polar code into multiple sub-blocks that have a same block length. The polar code may be partitioned in accordance with a modulation order of a modulation scheme (such as a QAM scheme), a quantity of bit channels, and different reliabilities associated with the bit channels. Further, the transmitting wireless communication device may assign information bits of an information bit vector to the bit channels using the multiple sub-blocks to generate multiple code blocks. The transmitting wireless communication device may interleave each code block with a respective bit-interleaver to obtain a codeword for transmission over the bit channels.

[0077] FIG. 2 shows an example of a wireless communications system **200** that supports polar coding techniques for higher-order modulation schemes. In some examples, the wireless communications system **200** may implement or be implemented by aspects of the wireless communications system **100**. For example, the wireless communications system **200** may include one or more transmitting wireless communication devices (such as a UE **115-a**) and one or more receiving wireless communication devices (such as a network entity **105-a**), which may be examples of the corresponding devices as described herein.

[0078] The UE **115-a** may include a memory, an encoder, and a transmitter. In some examples, the UE **115-a** may have data stored in the memory to transmit to another wireless communication device, such as the network entity **105-a**. To initiate the transmission process, the UE **115-a** may retrieve the data for transmission from the memory. The data may include a quantity of payload bits, ‘A,’ which may be 1s or 0s, provided from the memory to the encoder. In some examples, these payload bits may be combined with a quantity (such as a number) of parity or error checking bits, ‘E,’ to form a total set of information bits, ‘A+E.’ The quantity of information bits may be represented as a value ‘K,’ as shown. The encoder may implement a polar code with a block length, ‘N,’ for encoding the information bits. In some examples, N may be different than or the same as K. Such a polar code may be referred to as an (N, K) polar code. In some examples, the bits that are not allocated as information bits (such as N–K bits) may be assigned as frozen bits.

[0079] In some examples, to perform a polar coding operation, the encoder may generate a codeword of length, ‘M.’ In some examples, M may be a power of 2 (that is $M=2^{\sup.m}$, where m is an integer value). In some examples, if N is not a power of 2, the encoder may round the value of N up to a nearest valid M value. The encoder may implement an $(N=2^{\sup.m}, K)$ polar code according to Equation 1:

$$[00001] \quad c = u \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{\text{Math. } m} \quad (1)$$

where bits $u_{\text{sub}.i}$ are either information (that is, carrying payload) or frozen (such as $u_{\text{sub}.i}=0$). The matrix

$$[00002] \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{\text{Math. } m}$$

polarizes $2^{\sup.m}$ copies of a channel W into subchannels $W_{\text{sup}.i}$ that are either almost noisy (such as $I(W_{\text{sup}.i})_{\text{fwd}.0}$) or almost noiseless (such as $I(W_{\text{sup}.i})_{\text{fwd}.1}$). The information bits, $i_{\text{sub}.0}, \dots, i_{\text{sub}.K-1}$, may correspond to the K largest capacities $I(W_{\text{sup}.i})$. For example, if $N=400$, the encoder may determine a codeword length of $M=512$ (such as the nearest valid value for M greater than or equal to N) to support polar coding. In these examples, the encoder may encode a codeword of length M, and then may puncture a quantity of bits $M-N$ to obtain a codeword of the specified block length N for transmission.

[0080] In some examples, the encoder may use various bit-channel selection techniques to assign the information bits to the K most reliable bit channels, and the frozen bits may be assigned to the remaining bit channels. As an example (such as for large values of M or N, such as $M=1024$), the encoder may implement a FRActally enhanced Kernel (FRANK) polar code construction (which may be referred to as a mutual information-based polarization (MIP) scheme) for assigning the information bits K to the most (or an estimation of the most) reliable bit channels. To determine the distribution of information bits between bit channels or channel groups, the encoder may utilize mutual information. Mutual information may be an example of a metric for polarization, and may map rate distributions R for unpolarized channels W to rate distributions R1 and R2 for polarized channels $W_{\text{sup}.-}$ and $W_{\text{sup}.+}$, respectively. In some examples, an encoder may utilize mutual information (as opposed to reliability metrics) to assign bit channels in accordance with coding rate, R.

[0081] To construct a capacity-achieving polar code with successive cancellation (SC) decoding for a channel W with a coding rate

$$[00003] R = \frac{K}{N},$$

the encoder may distribute the information bits K such that the sub-code also achieve capacity on the polarized channels $W_{\text{sup}.-}$ and $W_{\text{sup}.+}$. To achieve capacity on channels $W_{\text{sup}.-}$ and $W_{\text{sup}.+}$ (for example, a polar transformation may take two copies of a channel W, resulting in channels $W_{\text{sup}.-}$ and $W_{\text{sup}.+}$, while preserving a capacity, I, of the channel W), the encoder may distribute information bits according to Equation 2:

$$[00004] K^- = I^- * (\frac{N}{2}), \text{ and } K^+ = I^+ * (\frac{N}{2}), \quad (2)$$

where $K.\text{sup.}-$ is a quantity of information bits to include in the first half of sub-blocks, $K.\text{sup.}+$ is a quantity of information bits to include in the second half of sub-blocks and $K = K.\text{sup.}+ + K.\text{sup.}-$. In some examples, $K.\text{sup.}-$ may be referred to as $K.\text{sub.upper}$ and $K.\text{sup.}+$ may be referred to as $K.\text{sub.lower}$. Further, $I.\text{sup.}-$ (such as $I.\text{sub.awgn.sup.}-$) is a capacity of the channel $W.\text{sup.}-$ and $I.\text{sup.}+$ (such as $I.\text{sub.awgn.sup.}+$) is a capacity of the channel $W.\text{sup.}+$ and may be calculated according to Equation 3:

$$[00005] I_{\text{awgn}} = 2I - I^2 - \delta, \text{ and } I_{\text{awgn}}^+ = I^2 + \delta \quad (3)$$

where I is a capacity of the channel W and δ is calculated according to Equation 4:

$$[00006] \delta = -\frac{\text{Math. } I - 0.5 \cdot \text{Math. } I}{32} + \frac{1}{64} \quad (4)$$

where $2I = I.\text{sup.}- + I.\text{sup.}+$. For FRANK polar code construction with greater than two channels, the encoder may recursively perform these polarizations and information bit assignment procedures. The encoder may stop recursively perform these polarizations and information bit assignment procedures in accordance with the block length, N , satisfying a threshold value (such as until a fixed length for which the encoder may know the information bit positions in accordance with an NR reliability sequence).

[0082] In some examples (such as for polar codes), the encoder may round the block length up to the nearest power of 2 to generate M coded bits. N coded bits may be transmitted on N unpunctured bit channels. In some examples, the encoder may perform rate matching for a relatively longer codeword to generate a codeword of a desired length. Rate matching may include puncturing (such as for relatively low target code rates) or shortening (such as for relatively high target code rates). For example, in examples in which the original code block is slightly greater than a power of 2, the encoder may puncture some quantity of coded bits (such as bits at the beginning of the codeword) or shorten some quantity of coded bits (such as bits at the end of the codeword). Here, the encoder may puncture a first quantity of coded bits, equal to $N-M$, of a set of information bits for relatively low rates or may shorten a last quantity of coded bits, equal to $N-M$, of the set of information bits for relatively high rates. As an example, if $M=1088$ and $N=2048$, then $N-M=960$ bits that may be punctured or shortened.

[0083] FRANK polar code construction may provide improved reliability for generated codewords than some other bit channel selection techniques (such as polarization weight (PW), generator weight (GW)), and in some examples may be relatively less complex than some other polar coding schemes (such as density evolution (DE)). Additionally, FRANK polar code construction may allow the encoder to flexibly adapt the coding rate when generating codewords. The encoder may determine information bit channels in accordance with FRANK polar code construction, and may assign frozen bits to the remaining channels. Frozen bits may be bits of a default value (such as 0 or 1) known to both the encoder and a decoder (such as the encoder that encodes information bits at a transmitter and the decoder that decodes the codeword received at a receiver). Further, the network entity **105-b** may receive a data signal representing the codeword via a receiver, and may decode the signal using a decoder to obtain the transmitted data. However, bits of QAM symbols associated with the different bit channels may have different reliabilities, and some techniques may not consider the different reliabilities (that is, some techniques may average the reliabilities).

[0084] Techniques described herein may support the consideration of different reliabilities for polar encoding of higher-order modulation schemes (such as QAM and others). That is, a transmitting wireless communication device, such as the UE **115-a**, may use multiple bit channels for the transmission of a codeword encoded using a polar code. Each bit channel may be associated with a different reliability (that is, the bit channels may be non-identical channels). For the purposes of this example, a quantity of the multiple bit channels may be 4, such that the bit channels may be represented by $W.\text{sub.0}$, $W.\text{sub.1}$, $W.\text{sub.2}$, and $W.\text{sub.3}$. Additionally, the UE **115-a** (such as an

encoder at the UE **115-a**) may apply a polar transform, A.sub.2, such as a 4×4 polar transform, to a polar code of total block length N to partition the polar code into multiple sub-blocks. Each sub-block has a same length equal to the total block length divided by the quantity of the bit channels (such as, N/4). The polar transform may be associated with a modulation order, the quantity of the multiple bit channels, and the different reliabilities associated with the multiple bit channels. In other words, partitioning the polar code into the multiple sub-blocks may result in multiple independent polar codes (such as independent code designs for each bit channel), such as 4 independent polar codes with length N/4. The UE **115-a** (such as the encoder at the UE **115-a**) may implement the sub-blocks (such as polar code sub-blocks), such that each sub-block polarizes a respective bit channel of the multiple bit channels into subchannels (such as W.sup.(--), W.sup.(+-), W.sup.(+-), and W.sup.(++)). The bit channels may be combined according to a method that results in a relatively fastest polarization (for example, as compared to other bit channel combinations).

[0085] Additionally, the UE **115-a** may assign information bits of an information bit vector to the bit channels using the multiple sub-blocks to generate multiple code blocks, corresponding to bit locations represented by Bit 1, Bit 2, Bit 3, and Bit 4, in FIG. 3. The UE **115-a** may identify a quantity of information bits of the information bit vector allocated to each sub-block in accordance with reliabilities respective associated with the quantity of information bits.

[0086] In some examples, the UE **115-a** may allocate the respective quantities of information bits to each sub-block via implementation of DE (such as DE for QAM (DE-Q)). For each sub-block, the UE **115-a** may calculate single-bit reliabilities of bit-channel sub-blocks for a given sub-block and may select a subset of the bit-channel sub-blocks for the given sub-block in accordance with the single-bit reliabilities. For example, the UE **115-a** may select K most reliable (for example, the best) bit-channel sub-blocks and may divide a set of information bits allocated to the given sub-block into partitions allocated to the subset of bit-channel sub-blocks for the given sub-block. Further, the UE **115-a** may determine information bit positions for each information bit of the set of information bits for the given sub-block based in accordance with the single-bit reliabilities.

[0087] Additionally, or alternatively, the UE **115-a** may allocate the respective quantities of information bits to each sub-block via implementation of FRANK (such as Finite Block Length (FBL)-MIP for QAM (FBL-MIP-Q)), as described previously. That is, the UE **115-a** may determine a rate distribution associated with each bit channel (such as R.sup.(--), R.sup.(+-), R.sup.(+-), and R.sup.(++)) associated with W.sup.(--), W.sup.(+-), W.sup.(+-), and W.sup.(++)), respectively) via FBL-FRANK and may implement MIP to recursively determine information bits for each sub-block (such as each N/4 sub-block). In some examples, the UE **115-a** may calculate the rate distributions (such as rate adjustments) according to Equation 5:

$$[00007] R^{(i)} \approx I^{(i)} - 0.88\left(\frac{N}{4}\right)^{-0.2438} \cdot \text{Math.} (I^{(i)} (1 - I^{(i)}))^{0.6741} \cdot \text{Math.} Q^{-1}(t) \quad (5)$$

where Q.sup.-1(t) is an inverse of a Gaussian Q function, and t is selected such that R.sup.(--)+R.sup.(+-)+R.sup.(+-)+R.sup.(++)=4R. N/4 may represent the same length of the sub-blocks. Additionally, the UE **115-a** may determine information bit positions for each quantity of information bits allocated to each sub-block (for example, equivalent to a code design for QPSK).

[0088] Additionally, the UE **115-a** may interleave each code block with a respective interleaver that may be referred to as a sub-block interleaver. In other words, each sub-block may be associated with a separate interleaver. For example, the interleavers associated with the code blocks may be triangle interleavers associated with cyclic shifts or random shifts, such that each interleaver is associated with a different cyclic shift or different random shift. In some examples, prior to interleaving the code blocks with the sub-block interleavers, the UE **115-a** may interleave each code block with row-in column-out interleavers with depth 2.sup.μ to support bit-to-constellation in non-rate-matched scenarios (such as similar to systematic bit priority mapping (SBPM) interleavers for low-density parity check (LDPC) codes). Conversely, for rate-matched scenarios, the UE **115-a**

may zero-pad a received sequence (for example, in a scenario of shortening) prior to interleaving the code blocks with the sub-block interleavers.

[0089] The UE **115-a** may transmit, to a receiving wireless communication device, such as the network entity **105-a**, a codeword **205** containing the multiple code blocks. That is, the UE **115-a** may transmit the respective interleaved bits of each code block over a respective bit channel. For example, the UE **115-a** may transmit Bit 2 over channel W.sub.2, Bit 3 over channel W.sub.3, Bit 1 over channel W.sub.1, and Bit 0 over channel W.sub.0.

[0090] While various aspects are described in the context of four bit channels, this is not to be regarded as a limitation of the present disclosure. In this regard, any quantity of bit channels associated with any quantity of polar code sub-blocks may be considered with reference to the techniques described herein. Further, although various aspects are described in the context of the UE **115-a** and the network entity **105-a**, this is not to be regarded as a limitation of the present disclosure. The UE **115-a** serves as an example of a transmitting wireless communication device and the network entity **105-a** serves as an example of a receiving wireless communication device, and any wireless communication device may be considered a transmitting wireless communication device or a receiving wireless communication device.

[0091] FIG. **3** shows an example of a rate matching scheme **300** that supports polar coding techniques for higher-order modulation schemes. In some examples, the rate matching scheme **300** may implement or be implemented by aspects of the wireless communications system **100**, the wireless communications system **200**, or both. For example, the rate matching scheme **300** may be implemented by one or more wireless communication devices, such as by one or more UEs **115**, one or more network entities **105**, or both, which may be examples of the corresponding devices as described herein.

[0092] In some examples, as described previously, an encoder of a UE **115** may round up the block length up to the nearest power of 2 to generate M coded bits **310**. In other words, the UE **115** may round up a $(N=2^{\text{sup.m}}, K)$ mother code to a (M, K) rate-matched code. As such, the encoder may shorten or puncture a quantity of coded bits **310** in each polar code sub-block **305**. In some examples, the quantity of coded bits is equal to a difference between a total block length of the polar code and a length of a codeword including the M coded bits **310** divided by a quantity of bit channels associated used by the UE **115** to transmit the code blocks. That is, continuing with the examples described with reference to FIG. **2** (such as shortening for 256 QAM), the UE **115-a** may partition a polar code of total block length N into multiple sub-blocks **305**. In some examples, each sub-block **305** has a length of $N/4$. As such, the encoder may shorten a last

[00008] $\frac{N-M}{4}$

coded bits **310** in each sub-block **305** or puncture a first

[00009] $\frac{N-M}{4}$

coded bits **310** in each sub-block **305**. For example, the encoder may partition a polar code into a sub-block **305-a**, a sub-block **305-b**, a sub-block **305-c**, and a sub-block **305-d**. Additionally, the encoder may shorten coded bits **310-a** in the sub-block **305-a**, coded bits **310-b** in the sub-block **305-b**, coded bits **310-c** in the sub-block **305-c**, and coded bits **310-d** in sub-block **305-d**.

[0093] For each sub-block **305**, information bit selection may be equivalent to a rate-matched polar code design for QPSK. As such, the UE **115-a** may support existing polar code construction methods.

[0094] FIG. **4** shows examples of rate matching schemes **400** (such as a rate matching scheme **400-a** and a rate matching scheme **400-b**) that supports polar coding techniques for higher-order modulation schemes. In some examples, the rate matching schemes **400** may implement or be implemented by aspects of the wireless communications system **100**, the wireless communications system **200**, the rate matching scheme **300**, or any combination thereof. For example, the rate matching schemes **400-a** and **400-b** may be implemented by one or more wireless communication devices, such as by one or more UEs **115**, one or more network entities **105**, or both, which may be

examples of the corresponding devices as described herein.

[0095] In some examples, a UE **115** may support multiple modulation schemes, as such, the UE **115** may support multiple rate matching schemes **400**. For example, the UE **115** may support a rate matching scheme **400-a**, which may be associated with a first modulation scheme. In some examples, the first modulation scheme is associated with $2^{\lceil \log_2 \mu \rceil}$ QAM symbols (for example, identical to 256 QAM symbols). In such examples, the UE **115** may use $2^{\lceil \log_2 \mu \rceil} - 1$ different bit channels (such as $W_{\text{sup},0}$ to $W_{\text{sup},(2^{\lceil \log_2 \mu \rceil} - 1)}$ bit channels) for the transmission of a codeword. Each sub-block **405** may be of a length

$$[00010] \frac{N}{2^{\lceil \log_2 \mu \rceil}}$$

and the UE **115** may apply a polar transform **410** (such as $A_{\text{sub},\mu-1}$). In some examples, the UE **115** may shorten a last

$$[00011] \frac{N-M}{2^{\lceil \log_2 \mu \rceil}}$$

coded bits **415** in each sub-block **405**. Additionally, or alternatively, the UE **115** may puncture a first

$$[00012] \frac{N-M}{2^{\lceil \log_2 \mu \rceil}}$$

coded bits in each sub-block.

[0096] In another example, the UE **115** may support a rate matching scheme **400-b**, which may be associated with a second modulation scheme. In some examples, the second modulation scheme is associated with $2^{\lceil \log_2 v \rceil}$ QAM symbols (such as when v is not a power of 2). In such examples, the UE **115** may add, to a codeword, $2^{\lceil \log_2 v \rceil} - v$ levels associated with a capacity 0, corresponding to blocks being punctured, or a capacity 1, corresponding to blocks being shortened. That is, each sub-block **405** may be of a length

$$[00013] \frac{N}{2^{\lceil \log_2 v \rceil}},$$

where $\mu - 1 = \lceil \log_2 v \rceil$, and the UE **115** may apply a polar transform **410** (such as $A_{\text{sub},\mu-1}$). The UE **115** may shorten a last

$$[00014] \frac{N-M}{2^{\lceil \log_2 v \rceil}}$$

coded bits each sub-block **405**. In some examples, the UE **115** may puncture a first

$$[00015] \frac{N-M}{2^{\lceil \log_2 v \rceil}}$$

coded bits in each sub-block **405** to support rate matching. Additionally, the UE **115** may add $2^{\lceil \log_2 v \rceil} - v$ levels associated with a capacity of 1 and shorten coded bits **415** (such as all coded bits) associated with the capacity of 1. In some examples, the UE **115** may add $2^{\lceil \log_2 v \rceil} - v$ levels associated with a capacity of 0 and shorten coded bits **415** (such as all coded bits) associated with the capacity of 0.

[0097] FIG. 5 shows an example of a process flow **500** that supports polar coding techniques for higher-order modulation schemes in accordance with one or more aspects of the present disclosure. In some examples, the process flow **500** may implement or be implemented by aspects of the wireless communications system **100**, the wireless communications system **200**, the rate matching scheme **300**, the rate matching schemes **400**, or any combination thereof. For example, the process flow **500** may include one or more transmitting wireless communication devices (such as a UE **115-b**) and one or more receiving wireless communication devices (such as a network entity **105-b**), which may be examples of the corresponding devices as described herein.

[0098] At **505**, the UE **115-b** may encode an information bit vector into multiple code blocks using a polar code that is partitioned into multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In such examples, each code block of the multiple code blocks may be encoded using a respective sub-block of the multiple sub-blocks. Additionally, the same block length of each sub-block of the multiple sub-blocks may be equal to a total block length of the polar code, N , divided by a quantity of bit channels used by the UE **115-b** to transmit the code blocks.

[0099] In some examples, encoding the information bit vector may include, at **510**, allocating a

respective quantity of bits of the information bit vector to each sub-block of the multiple sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector. That is, the UE **115-b** may be associated with multiple bit channels (such as the quantity of bit channels) and each bit channel may be associated with a different reliability. Additionally, each sub-block may be associated with a bit channel of the multiple bit channels. As such, the UE **115-b** may allocate the respective quantity of bits of the information bit vector to each sub-block of the multiple sub-blocks in accordance with the reliabilities respectively associated with the bits of the information bit vector and the different reliabilities associated with the bit channels of the multiple bit channels. In other words, bits of the information bit vector associated with a greater reliability (for example, out of the reliabilities respectively associated with the bits of the information bit vector) may be allocated to a sub-block associated with a bit channel that is further associated with a greater reliability (for example, out of the different reliabilities associated with the multiple bit channels)

[0100] In some examples (such as in examples where the UE **115-b** allocates bits according to DE-Q), to allocate the respective quantity of bits of the information bit vector to each sub-block of the multiple sub-blocks, the UE **115-b** may determine single-bit reliabilities of respective bit-channel sub-blocks for each sub-block of the multiple sub-blocks and may select, for each sub-block of the multiple sub-blocks, respective subsets of the respective bit-channel sub-blocks in accordance with the single-bit reliabilities. That is, for each sub-block, the UE **115-b** may select K most reliable (such as associated with the K greatest single-bit reliabilities) bit channel sub-blocks of the respective bit-channel sub-blocks. As such, the UE **115-b** may divide respective sets of bits into partitions allocated to the respective subsets of the respective bit channel sub-blocks for each sub-block in accordance with the single-bit reliabilities.

[0101] In some other examples (such as in examples when the UE **115-b** allocates bits according to FBL-MIP-Q), to allocate the respective quantity of bits of the information bit vector to each sub-block of the multiple sub-blocks, the UE **115-b** may divide respective sets of bits into partitions allocated to bit-channel sub-blocks for each sub-block of the multiple sub-blocks in accordance with a mutual information transfer function. As such, the respective quantity of bits of the information bit vector may be allocated to each sub-block of the multiple sub-blocks in accordance with the mutual information transfer function. In some examples, the respective sets of bits may be divided recursively, such that the recursive dividing is stopped in accordance with the bit channel sub-blocks being associated with a threshold length. The threshold length may be a fixed length for which the UE **115-b** may know the information bit positions in accordance with an NR reliability sequence.

[0102] In some examples, at **515**, the UE **115-b** may determine a bit position for each bit of the respective quantity of bits allocated to each sub-block of the multiple sub-blocks. As such, each code block of the multiple code blocks may include the respective quantity of bits of the information bit vector in accordance with the bit position. In some examples where the UE **115-b** allocates bits according to DE-Q, the UE **115-b** may determine the bit position for each bit of the partitions in each bit channel sub-block in accordance with the single-bit reliabilities. In some examples where the UE **115-b** allocates bits according to FBL-MIP-Q, the UE **115-b** may determine the bit position for each bit of the respective quantity of bits in accordance with the bit channel sub-blocks being within the threshold length. In some examples, at **520**, the UE **115-b** may map the bits of the information bit vector to respective bit locations associated with the multiple sub-blocks.

[0103] In some examples, at **525**, the UE **115-b** may perform rate matching for the respective quantity of encoded bits associated with each code block in accordance with the same block length of each sub-block. For example, at **530**, the UE **115-b** may shorten a first subset of the respective quantity of the encoded bits associated with each code block. In some examples, the first subset includes a first quantity of the encoded bits at an end of the respective quantity of the encoded bits

associated with each code block. The first quantity of encoded bits may be equal to a difference between the total block length of the polar code (such as N) and a length of a codeword (such as M) divided by the quantity of bit channels used by the UE **115-b** to transmit the code blocks.

[0104] Additionally, or alternatively, the UE **115-b** may shorten the first subset of the respective quantity of the encoded bits associated with one or more code blocks of the multiple code blocks and may shorten all of the encoded bits associated with one or more other code blocks from the multiple code blocks. In some examples, the one or more other code blocks may be associated with a capacity level of 1.

[0105] In another example, at **535**, the UE **115-b** may puncture a second subset of the respective quantity of the encoded bits associated with each code block, the second subset comprising a second quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with each code block. Similarly, the second quantity of encoded bits may be equal to the difference between the total block length of the polar code (such as N) and the length of a codeword (such as M) divided by the quantity of bit channels used by the UE **115-b** to transmit the code blocks.

[0106] Additionally, or alternatively, the UE **115-b** may puncture the second subset of the respective quantity of the encoded bits associated with one or more code blocks of the multiple code blocks and may puncture all of the encoded bits associated with one or more other code blocks from the multiple code blocks. In some examples, the one or more other code blocks may be associated with a capacity level of 0.

[0107] In some examples, at **540**, the UE **115-b** may map the encoded bits to a constellation in accordance with a row-in column-out interleaving scheme, the encoded bits being mapped to the constellation prior to interleaving each code block.

[0108] At **545**, the UE **115-b** may interleave the respective bits of each code block of the multiple code blocks using a respective interleaver of multiple interleavers. That is, each interleaver of the multiple interleavers may be associated with different code blocks of the multiple code blocks, such that each interleaver is different. For example, the multiple interleavers may be triangle interleavers, such that each interleaver is associated with a different cyclic or random shift. The bits of each code block may be interleaved in accordance with the respective bit locations.

[0109] At **550**, the UE **115-b** may transmit, to the network entity **105-b**, the respective interleaved bits of each code block of the multiple code blocks over a respective bit channel of the multiple bit channels. That is, the network entity **105-b** may receive (for example, obtain) a signal representing the multiple code blocks (such as including the information bit vector) over the multiple bit channels.

[0110] At **555**, the network entity **105-b** may de-interleave each code block of the multiple code blocks using multiple de-interleavers. In some examples, each de-interleaver is associated with a respective code block of the multiple code blocks. That is, the network entity **105-b** may de-interleave each code block using multiple de-interleavers in accordance with each code block being interleaved with a different interleaver.

[0111] At **560**, the network entity **105-b** may decode the multiple code blocks according to the polar code (for example, according to the multiple sub-blocks) to obtain the information bit vector.

[0112] Although some examples are described in the context of the UE **115-b** and the network entity **105-b**, this is not to be regarded as a limitation of the present disclosure. In this regard, the UE **115-b** serves as an example of a transmitting wireless communication device and the network entity **105-b** serves as an example of a receiving wireless communication device, such that any wireless communication device may be considered a transmitting wireless communication device or a receiving wireless communication device.

[0113] FIG. **6** shows a block diagram of an example wireless communication device **605** that supports polar coding techniques for higher-order modulation schemes. The wireless communication device **605** may be an example of aspects of a UE **115** or a network entity **105** as

described herein. The wireless communication device **605** may include a receiver **610**, a transmitter **615**, and a communications manager **620**. The wireless communication device **605**, or one or more components of the wireless communication device **605** (such as the receiver **610**, the transmitter **615**, and the communications manager **620**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (such as via one or more buses).

[0114] The receiver **610** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (such as control channels, data channels, information channels related to polar coding techniques for higher-order modulation). Information may be passed on to other components of the wireless communication device **605**. The receiver **610** may utilize a single antenna or a set of multiple antennas.

[0115] The transmitter **615** may provide a means for transmitting signals generated by other components of the wireless communication device **605**. For example, the transmitter **615** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (such as control channels, data channels, information channels related to polar coding techniques for higher-order modulation). In some examples, the transmitter **615** may be co-located with a receiver **610** in a transceiver module. The transmitter **615** may utilize a single antenna or a set of multiple antennas.

[0116] The communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations thereof or various components thereof may be examples of means for performing various aspects of polar coding techniques for higher-order modulation as described herein. For example, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0117] In some examples, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in hardware (such as in communications management circuitry). The hardware may include at least one of a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (such as by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0118] Additionally, or alternatively, the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be implemented in code (such as communications management software or firmware) executed by at least one processor. If implemented in code executed by at least one processor, the functions of the communications manager **620**, the receiver **610**, the transmitter **615**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (for example, configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0119] In some examples, the communications manager **620** may be configured to perform various operations (such as receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **610**, the transmitter **615**, or both. For example, the communications manager **620** may receive information from the receiver **610**, send information to the transmitter

615, or be integrated in combination with the receiver **610**, the transmitter **615**, or both to obtain information, output information, or perform various other operations as described herein.

[0120] The communications manager **620** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **620** is capable of, configured to, or operable to support a means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The communications manager **620** is capable of, configured to, or operable to support a means for interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers. The communications manager **620** is capable of, configured to, or operable to support a means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0121] Additionally, or alternatively, the communications manager **620** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **620** is capable of, configured to, or operable to support a means for receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The communications manager **620** is capable of, configured to, or operable to support a means for de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The communications manager **620** is capable of, configured to, or operable to support a means for decoding the set of multiple code blocks according to the polar code to obtain the information bit vector.

[0122] By including or configuring the communications manager **620** in accordance with examples as described herein, the wireless communication device **605** (such as at least one processor controlling or otherwise coupled with the receiver **610**, the transmitter **615**, the communications manager **620**, or a combination thereof) may support techniques for partitioning a polar code into multiple sub-blocks for encoding of an information bit vector, which may result in reduced processing, reduced power consumption, and more efficient utilization of communication resources, among other advantages.

[0123] FIG. 7 shows a block diagram of an example wireless communication device **705** that supports polar coding techniques for higher-order modulation schemes. The wireless communication device **705** may be an example of aspects of a wireless communication device **605**, a UE **115**, or a network entity **105** as described herein. The wireless communication device **705** may include a receiver **710**, a transmitter **715**, and a communications manager **720**. The wireless communication device **705**, or one of more components of the wireless communication device **705** (such as the receiver **710**, the transmitter **715**, and the communications manager **720**), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (such as via one or more buses). The communications manager **720** can be implemented, at least in part, by one or both of a modem and a processor.

[0124] The receiver **710** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (such as control channels, data channels, information channels related to polar coding techniques for

higher-order modulation schemes). Information may be passed on to other components of the wireless communication device **705**. The receiver **710** may utilize a single antenna or a set of multiple antennas.

[0125] The transmitter **715** may provide a means for transmitting signals generated by other components of the wireless communication device **705**. For example, the transmitter **715** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (such as control channels, data channels, information channels related to polar coding techniques for higher-order modulation schemes). In some examples, the transmitter **715** may be co-located with a receiver **710** in a transceiver module. The transmitter **715** may utilize a single antenna or a set of multiple antennas.

[0126] The wireless communication device **705**, or various components thereof, may be an example of means for performing various aspects of polar coding techniques for higher-order modulation as described herein. For example, the communications manager **720** may include an encoding component **725**, an interleaving component **730**, a bit channel component **735**, a de-interleaving component **740**, a decoding component **745**, or any combination thereof. The communications manager **720** may be an example of aspects of a communications manager **620** as described herein. In some examples, the communications manager **720**, or various components thereof, may be configured to perform various operations (such as receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **710**, the transmitter **715**, or both. For example, the communications manager **720** may receive information from the receiver **710**, send information to the transmitter **715**, or be integrated in combination with the receiver **710**, the transmitter **715**, or both to obtain information, output information, or perform various other operations as described herein.

[0127] The communications manager **720** may support wireless communications in accordance with examples as disclosed herein. The encoding component **725** is capable of, configured to, or operable to support a means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The interleaving component **730** is capable of, configured to, or operable to support a means for interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers. The bit channel component **735** is capable of, configured to, or operable to support a means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0128] Additionally, or alternatively, the communications manager **720** may support wireless communications in accordance with examples as disclosed herein. The bit channel component **735** is capable of, configured to, or operable to support a means for receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The de-interleaving component **740** is capable of, configured to, or operable to support a means for de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The decoding component **745** is capable of, configured to, or operable to support a means for decoding the set of multiple code blocks according to the polar code to obtain the information bit vector.

[0129] FIG. 8 shows a block diagram of an example communications manager **820** that supports polar coding techniques for higher-order modulation schemes. The communications manager **820**, or various components thereof, may be an example of means for performing various aspects of polar coding techniques for higher-order modulation schemes as described herein. For example, the communications manager **820** may include an encoding component **825**, an interleaving component **830**, a bit channel component **835**, a de-interleaving component **840**, a decoding component **845**, a rate matching component **850**, a shortening component **855**, a puncturing component **860**, or any combination thereof. Each of these components, or components or subcomponents thereof (such as one or more processors, one or more memories), may communicate, directly or indirectly, with one another (such as via one or more buses), which may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (such as between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity **105**, between devices, components, or virtualized components associated with a network entity **105**), or any combination thereof.

[0130] The communications manager **820** may support wireless communications in accordance with examples as disclosed herein. The encoding component **825** is capable of, configured to, or operable to support a means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The interleaving component **830** is capable of, configured to, or operable to support a means for interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers. The bit channel component **835** is capable of, configured to, or operable to support a means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0131] In some examples, to support encoding the information bit vector, the encoding component **825** is capable of, configured to, or operable to support a means for mapping bits of the information bit vector to respective bit locations associated with the set of multiple sub-blocks,. In some examples, the bits of each code block are interleaved in accordance with the respective bit locations.

[0132] In some examples, to support encoding the information bit vector, the encoding component **825** is capable of, configured to, or operable to support a means for allocating a respective quantity of bits of the information bit vector to each sub-block of the set of multiple sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector. In some examples, to support encoding the information bit vector, the encoding component **825** is capable of, configured to, or operable to support a means for determining a bit position for each bit of the respective quantity of bits allocated to each sub-block of the set of multiple sub-blocks. In some examples, each code block of the set of multiple code blocks includes the respective quantity of bits of the information bit vector in accordance with the bit positions.

[0133] In some examples, to support allocating the quantities of bits of the information bit vector to the set of multiple sub-blocks, the encoding component **825** is capable of, configured to, or operable to support a means for calculating single-bit reliabilities of respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks. In some examples, to support allocating the quantities of bits of the information bit vector to the set of multiple sub-blocks, the encoding component **825** is capable of, configured to, or operable to support a means for selecting respective subsets of the respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with the single-bit reliabilities. In some examples, to support allocating the quantities of bits of the information bit vector to the set of multiple sub-blocks, the encoding component **825**

is capable of, configured to, or operable to support a means for dividing respective sets of bits into partitions allocated to the respective subsets of the respective bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with the single-bit reliabilities.

[0134] In some examples, to support determining the bit position for each bit of the respective quantity of bits, the encoding component **825** is capable of, configured to, or operable to support a means for determining the bit position for each bit of the partitions in each bit channel sub-block in accordance with the single-bit reliabilities.

[0135] In some examples, to support allocating the respective quantity of bits of the information bit vector to each sub-block of the set of multiple sub-blocks, the encoding component **825** is capable of, configured to, or operable to support a means for dividing respective sets of bits into partitions allocated to bit channel sub-blocks for each sub-block of the set of multiple sub-blocks. In some examples, the respective sets of bits are divided in accordance with a mutual information transfer function. In some examples, the respective quantity of bits is allocated to each sub-block of the set of multiple sub-blocks in accordance with the mutual information transfer function.

[0136] In some examples, the respective sets of bits are divided into partitions recursively. In some examples, the recursive dividing is stopped based on the bit channel sub-blocks being associated with a threshold length. In some examples, determining the bit position for each bit of the respective quantity of bits is based on the bit channel sub-blocks being associated with the threshold length.

[0137] In some examples, each code block is associated with a respective quantity of encoded bits, and the rate matching component **850** is capable of, configured to, or operable to support a means for performing rate matching for the respective quantity of the encoded bits associated with each code block in accordance with the same block length of each sub-block.

[0138] In some examples, to support performing the rate matching for the respective quantity of the encoded bits, the shortening component **855** is capable of, configured to, or operable to support a means for shortening a subset of the respective quantity of the encoded bits associated with each code block, the subset including a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with each code block.

[0139] In some examples, the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword including the set of multiple code blocks divided by a quantity of the set of multiple bit channels.

[0140] In some examples, to support performing the rate matching for the respective quantity of the encoded bits, the puncturing component **860** is capable of, configured to, or operable to support a means for puncturing a subset of the respective quantity of the encoded bits associated with each code block, the subset including a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with each code block.

[0141] In some examples, the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword including the set of multiple code blocks divided by a quantity of the set of multiple bit channels.

[0142] In some examples, to support performing the rate matching for the respective quantity of the encoded bits, the shortening component **855** is capable of, configured to, or operable to support a means for shortening a subset of the respective quantity of the encoded bits associated with one or more code blocks from the set of multiple code blocks, the subset of the respective quantity of the encoded bits including a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with the one or more code blocks. In some examples, to support performing the rate matching for the respective quantity of the encoded bits, the shortening component **855** is capable of, configured to, or operable to support a means for shortening all of the encoded bits associated with one or more other code blocks from the set of multiple code blocks. In some examples, the one or more other code blocks are associated with a capacity level of 1.

[0143] In some examples, to support performing the rate matching for the respective quantity of the

encoded bits, the puncturing component **860** is capable of, configured to, or operable to support a means for puncturing a subset of the respective quantity of the encoded bits associated with one or more code blocks from the set of multiple code blocks, the subset of the respective quantity of the encoded bits including a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with the one or more code blocks. In some examples, to support performing the rate matching for the respective quantity of the encoded bits, the puncturing component **860** is capable of, configured to, or operable to support a means for puncturing all of the encoded bits associated with one or more other code blocks from the set of multiple code blocks. In some examples, the one or more other code blocks are associated with a capacity level of 0.

[0144] In some examples, a total block length of the polar code is equal to N . In some examples, the same block length of each sub-block of the set of multiple sub-blocks is equal to N divided by a quantity of the set of multiple bit channels.

[0145] In some examples, the encoding component **825** is capable of, configured to, or operable to support a means for mapping encoded bits to a constellation in accordance with a row-in column-out interleaving scheme, the encoded bits being mapped to the constellation prior to interleaving each code block using the respective interleaver of the set of multiple interleavers.

[0146] In some examples, each interleaver of the set of multiple interleavers is different in accordance with each interleaver being associated with different code blocks of the set of multiple code blocks.

[0147] Additionally, or alternatively, the communications manager **820** may support wireless communications in accordance with examples as disclosed herein. In some examples, the bit channel component **835** is capable of, configured to, or operable to support a means for receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The de-interleaving component **840** is capable of, configured to, or operable to support a means for de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The decoding component **845** is capable of, configured to, or operable to support a means for decoding the set of multiple code blocks according to the polar code to obtain the information bit vector.

[0148] In some examples, bits of the information bit vector are mapped to respective bit locations associated with the set of multiple sub-blocks. In some examples, a respective quantity of bits of the information bit vector is allocated to each sub-block of the set of multiple sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector. In some examples, each code block of the set of multiple code blocks includes the respective quantity of bits of the information bit vector in accordance with bit positions.

[0149] In some examples, respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with single-bit reliabilities respectively associated with the bit channel sub-blocks. In some examples, the bit channel sub-blocks for each sub-block are selected from a set of multiple bit channel sub-blocks in accordance with the single-bit reliabilities respectively associated with the bit channel sub-blocks.

[0150] In some examples, respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the set of multiple sub-blocks in accordance with a mutual information transfer function. In some examples, the respective quantity of the bits is allocated to each sub-block of the set of multiple sub-blocks in accordance with the mutual information transfer function.

[0151] In some examples, each code block is associated with a respective quantity of encoded bits, the respective quantity of the encoded bits being rate matched in accordance with the same block length of each sub-block.

[0152] In some examples, a subset of the respective quantity of the encoded bits associated with each code block is shortened, the subset including a first quantity of the encoded bits at an end of the respective quantity of the encoded bits.

[0153] In some examples, the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword including the set of multiple code blocks divided by a quantity of the set of multiple bit channels.

[0154] In some examples, a subset of the respective quantity of the encoded bits associated with each code block is punctured, the subset including a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits.

[0155] In some examples, the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword including the set of multiple code blocks divided by a quantity of the set of multiple bit channels.

[0156] In some examples, a subset of the respective quantity of the encoded bits for one or more code blocks from the set of multiple code blocks is shortened, the subset of the respective quantity of the encoded bits including a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks from the set of multiple code blocks are shortened. In some examples, the one or more other code blocks are associated with a capacity level of 1.

[0157] In some examples, a subset of the respective quantity of the encoded bits for one or more code blocks from the set of multiple code blocks is punctured, the subset of the respective quantity of the encoded bits including a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks of the set of multiple code blocks are punctured. In some examples, the one or more other code blocks are associated with a capacity level of 0.

[0158] In some examples, a total block length of the polar code is equal to N . In some examples, the same block length of each sub-block of the set of multiple sub-blocks is equal to N divided by a quantity of the set of multiple bit channels.

[0159] FIG. 9 shows a diagram of an example system including a wireless communication device **905** that supports polar coding techniques for higher-order modulation in accordance with one or more aspects of the present disclosure. The wireless communication device **905** may be an example of or include the components of a wireless communication device **605**, a wireless communication device **705**, or a UE **115** as described herein. The wireless communication device **905** may communicate (for example, wirelessly) with one or more network entities **105**, one or more UEs **115**, or any combination thereof. The wireless communication device **905** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **920**, an input/output (I/O) controller **910**, a transceiver **915**, an antenna **925**, at least one memory **930**, code **935**, and at least one processor **940**. These components may be in electronic communication or otherwise coupled (for example, operatively, communicatively, functionally, electronically, electrically) via one or more buses (such as a bus **945**).

[0160] The I/O controller **910** may manage input and output signals for the wireless communication device **905**. The I/O controller **910** may also manage peripherals not integrated into the wireless communication device **905**. In some examples, the I/O controller **910** may represent a physical connection or port to an external peripheral. In some examples, the I/O controller **910** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **910** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a

similar wireless communication device. In some examples, the I/O controller **910** may be implemented as part of one or more processors, such as the at least one processor **940**. In some examples, a user may interact with the wireless communication device **905** via the I/O controller **910** or via hardware components controlled by the I/O controller **910**.

[0161] In some examples, the wireless communication device **905** may include a single antenna **925**. However, in some other examples, the wireless communication device **905** may have more than one antenna **925**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **915** may communicate bi-directionally, via the one or more antennas **925**, wired, or wireless links as described herein. For example, the transceiver **915** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **915** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **925** for transmission, and to demodulate packets received from the one or more antennas **925**. The transceiver **915**, or the transceiver **915** and one or more antennas **925**, may be an example of a transmitter **615**, a transmitter **715**, a receiver **610**, a receiver **710**, or any combination thereof or component thereof, as described herein.

[0162] The at least one memory **930** may include random access memory (RAM) and read-only memory (ROM). The at least one memory **930** may store computer-readable, computer-executable code **935** including instructions that, when executed by the at least one processor **940**, cause the wireless communication device **905** to perform various functions described herein. The code **935** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some examples, the code **935** may not be directly executable by the at least one processor **940** but may cause a computer (for example, when compiled and executed) to perform functions described herein. In some examples, the at least one memory **930** may contain, among other things, a basic I/O system (BIOS), which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0163] The at least one processor **940** may include an intelligent hardware device (such as a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some examples, the at least one processor **940** may be configured to operate a memory array using a memory controller. In some other examples, a memory controller may be integrated into the at least one processor **940**. The at least one processor **940** may be configured to execute computer-readable instructions stored in a memory (such as the at least one memory **930**) to cause the wireless communication device **905** to perform various functions (such as functions or tasks supporting polar coding techniques for higher-order modulation). For example, the wireless communication device **905** or a component of the wireless communication device **905** may include at least one processor **940** and at least one memory **930** coupled with or to the at least one processor **940**, the at least one processor **940** and at least one memory **930** configured to perform various functions described herein. In some examples, the at least one processor **940** may include multiple processors and the at least one memory **930** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein.

[0164] The communications manager **920** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **920** is capable of, configured to, or operable to support a means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The communications manager **920** is capable of, configured to, or operable to support a means for interleaving respective bits of each code block of the set of

multiple code blocks using a respective interleaver of a set of multiple interleavers. The communications manager **920** is capable of, configured to, or operable to support a means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0165] Additionally, or alternatively, the communications manager **920** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **920** is capable of, configured to, or operable to support a means for receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The communications manager **920** is capable of, configured to, or operable to support a means for de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The communications manager **920** is capable of, configured to, or operable to support a means for decoding the set of multiple code blocks according to the polar code to obtain the information bit vector.

[0166] By including or configuring the communications manager **920** in accordance with examples as described herein, the wireless communication device **905** may support techniques partitioning a polar code into multiple sub-blocks for encoding of an information bit vector, which may result in improved communication reliability, reduced latency, improved user experience related to reduced processing, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, and improved utilization of processing capability, among other advantages.

[0167] In some examples, the communications manager **920** may be configured to perform various operations (such as receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **915**, the one or more antennas **925**, or any combination thereof. Although the communications manager **920** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **920** may be supported by or performed by the at least one processor **940**, the at least one memory **930**, the code **935**, or any combination thereof. For example, the code **935** may include instructions executable by the at least one processor **940** to cause the wireless communication device **905** to perform various aspects of polar coding techniques for higher-order modulation as described herein, or the at least one processor **940** and the at least one memory **930** may be otherwise configured to, individually or collectively, perform or support such operations.

[0168] FIG. **10** shows a diagram of an example system including a wireless communication device **1005** that supports polar coding techniques for higher-order modulation schemes. The wireless communication device **1005** may be an example of or include the components of a wireless communication device **605**, a wireless communication device **705**, or a network entity **105** as described herein. The wireless communication device **1005** may communicate with one or more network entities **105**, one or more UEs **115**, or any combination thereof, which may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The wireless communication device **1005** may include components that support outputting and obtaining communications, such as a communications manager **1020**, a transceiver **1010**, an antenna **1015**, at least one memory **1025**, code **1030**, and at least one processor **1035**. These components may be in electronic communication or otherwise coupled (for example, operatively, communicatively, functionally, electronically, electrically) via one or more buses (such as a bus **1040**).

[0169] The transceiver **1010** may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver **1010** may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver **1010** may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the wireless communication device **1005** may include one or more antennas **1015**, which may be capable of transmitting or receiving wireless transmissions (for example, concurrently). The transceiver **1010** may also include a modem to modulate signals, to provide the modulated signals for transmission (such as by one or more antennas **1015**, by a wired transmitter), to receive modulated signals (such as from one or more antennas **1015**, from a wired receiver), and to demodulate signals. In some implementations, the transceiver **1010** may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas **1015** that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas **1015** that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver **1010** may include or be configured for coupling with one or more processors or one or more memory components that are operable to perform or support operations based on received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver **1010**, or the transceiver **1010** and the one or more antennas **1015**, or the transceiver **1010** and the one or more antennas **1015** and one or more processors or one or more memory components (such as the at least one processor **1035**, the at least one memory **1025**, or both), may be included in a chip or chip assembly that is installed in the wireless communication device **1005**. In some examples, the transceiver **1010** may be operable to support communications via one or more communications links (such as a communication link **125**, a backhaul communication link **120**, a midhaul communication link **162**, a fronthaul communication link **168**).

[0170] The at least one memory **1025** may include RAM, ROM, or any combination thereof. The at least one memory **1025** may store computer-readable, computer-executable code **1030** including instructions that, when executed by one or more of the at least one processor **1035**, cause the wireless communication device **1005** to perform various functions described herein. The code **1030** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some examples, the code **1030** may not be directly executable by a processor of the at least one processor **1035** but may cause a computer (for example, when compiled and executed) to perform functions described herein. In some examples, the at least one memory **1025** may contain, among other things, a BIOS, which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some examples, the at least one processor **1035** may include multiple processors and the at least one memory **1025** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein (for example, as part of a processing system).

[0171] The at least one processor **1035** may include an intelligent hardware device (such as a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA, a microcontroller, a programmable logic device, discrete gate or transistor logic, a discrete hardware component, or any combination thereof). In some examples, the at least one processor **1035** may be configured to operate a memory array using a memory controller. In some other examples, a memory controller may be integrated into one or more of the at least one processor **1035**. The at least one processor **1035** may be configured to execute computer-readable instructions stored in a memory (such as one or more of the at least one memory **1025**) to cause the wireless communication device **1005** to perform various functions (such as functions or tasks supporting polar coding techniques for higher-order modulation). For example, the wireless communication device **1005** or a component of the wireless communication device **1005** may include at least one processor **1035** and at least one memory **1025**

coupled with one or more of the at least one processor **1035**, the at least one processor **1035** and the at least one memory **1025** configured to perform various functions described herein. The at least one processor **1035** may be an example of a cloud-computing platform (such as one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (for example, by executing code **1030**) to perform the functions of the wireless communication device **1005**. The at least one processor **1035** may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the wireless communication device **1005** (such as within one or more of the at least one memory **1025**). In some implementations, the at least one processor **1035** may be a component of a processing system. A processing system may generally refer to a system or series of machines or components that receives inputs and processes the inputs to produce a set of outputs (which may be passed to other systems or components of, for example, the wireless communication device **1005**). For example, a processing system of the wireless communication device **1005** may refer to a system including the various other components or subcomponents of the wireless communication device **1005**, such as the at least one processor **1035**, or the transceiver **1010**, or the communications manager **1020**, or other components or combinations of components of the wireless communication device **1005**. The processing system of the wireless communication device **1005** may interface with other components of the wireless communication device **1005**, and may process information received from other components (such as inputs or signals) or output information to other components. For example, a chip or modem of the wireless communication device **1005** may include a processing system and one or more interfaces to output information, or to obtain information, or both. The one or more interfaces may be implemented as or otherwise include a first interface configured to output information and a second interface configured to obtain information, or a same interface configured to output information and to obtain information, among other implementations. In some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a transmitter, such that the wireless communication device **1005** may transmit information output from the chip or modem. Additionally, or alternatively, in some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a receiver, such that the wireless communication device **1005** may obtain information or signal inputs, and the information may be passed to the processing system. A person having ordinary skill in the art will readily recognize that a first interface also may obtain information or signal inputs, and a second interface also may output information or signal outputs.

[0172] In some examples, a bus **1040** may support communications of (for example, within) a protocol layer of a protocol stack. In some examples, a bus **1040** may support communications associated with a logical channel of a protocol stack (for example, between protocol layers of a protocol stack), which may include communications performed within a component of the wireless communication device **1005**, or between different components of the wireless communication device **1005** that may be co-located or located in different locations (for example, where the wireless communication device **1005** may refer to a system in which one or more of the communications manager **1020**, the transceiver **1010**, the at least one memory **1025**, the code **1030**, and the at least one processor **1035** may be located in one of the different components or divided between different components).

[0173] In some examples, the communications manager **1020** may manage aspects of communications with a core network **130** (such as via one or more wired or wireless backhaul links). For example, the communications manager **1020** may manage the transfer of data communications for client devices, such as one or more UEs **115**. In some examples, the communications manager **1020** may manage communications with other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** in cooperation with other network entities **105**. In some examples, the communications manager **1020** may

support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**.

[0174] The communications manager **1020** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1020** is capable of, configured to, or operable to support a means for encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The communications manager **1020** is capable of, configured to, or operable to support a means for interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers. The communications manager **1020** is capable of, configured to, or operable to support a means for transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability.

[0175] Additionally, or alternatively, the communications manager **1020** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1020** is capable of, configured to, or operable to support a means for receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The communications manager **1020** is capable of, configured to, or operable to support a means for de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The communications manager **1020** is capable of, configured to, or operable to support a means for decoding the set of multiple code blocks according to the polar code to obtain the information bit vector.

[0176] By including or configuring the communications manager **1020** in accordance with examples as described herein, the wireless communication device **1005** may support techniques partitioning a polar code into multiple sub-blocks for encoding of an information bit vector, which may result in improved communication reliability, reduced latency, improved user experience related to reduced processing, reduced power consumption, more efficient utilization of communication resources, improved coordination between devices, longer battery life, and improved utilization of processing capability, among other advantages.

[0177] In some examples, the communications manager **1020** may be configured to perform various operations (such as receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **1010**, the one or more antennas **1015** (for example, where applicable), or any combination thereof. Although the communications manager **1020** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1020** may be supported by or performed by the transceiver **1010**, one or more of the at least one processor **1035**, one or more of the at least one memory **1025**, the code **1030**, or any combination thereof (for example, by a processing system including at least a portion of the at least one processor **1035**, the at least one memory **1025**, the code **1030**, or any combination thereof). For example, the code **1030** may include instructions executable by one or more of the at least one processor **1035** to cause the wireless communication device **1005** to perform various aspects of polar coding techniques for higher-order modulation as described herein, or the at least one processor **1035** and the at least one memory **1025** may be

otherwise configured to, individually or collectively, perform or support such operations.

[0178] FIG. 11 shows a flowchart illustrating a method 1100 that supports polar coding techniques for higher-order modulation in accordance with aspects of the present disclosure. The operations of the method 1100 may be implemented by a wireless communication device, such as a UE or a network entity, or its components as described herein. For example, the operations of the method 1100 may be performed by a UE 115 or a network entity as described with reference to FIGS. 1-10. In some examples, a UE or a network entity may execute a set of instructions to control the functional elements of the UE or the network entity to perform the described functions.

Additionally, or alternatively, the UE or the network entity may perform aspects of the described functions using special-purpose hardware.

[0179] At block 1105, the method may include encoding an information bit vector into a set of multiple code blocks using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the set of multiple code blocks being encoded using a respective sub-block of the set of multiple sub-blocks. The operations of block 1105 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of block 1105 may be performed by an encoding component 825 as described with reference to FIG. 8.

[0180] At block 1110, the method may include interleaving respective bits of each code block of the set of multiple code blocks using a respective interleaver of a set of multiple interleavers. The operations of block 1110 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of block 1110 may be performed by an interleaving component 830 as described with reference to FIG. 8.

[0181] At block 1115, the method may include transmitting the respective interleaved bits of each code block of the set of multiple code blocks over a respective bit channel of a set of multiple bit channels, each bit channel of the set of multiple bit channels being associated with a respective reliability. The operations of block 1115 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of block 1115 may be performed by a bit channel component 835 as described with reference to FIG. 8.

[0182] FIG. 12 shows a flowchart illustrating a method 1200 that supports polar coding techniques for higher-order modulation in accordance with aspects of the present disclosure. The operations of the method 1200 may be implemented by a wireless communication device, such as a UE or a network entity, or its components as described herein. For example, the operations of the method 1200 may be performed by a UE 115 or a network entity as described with reference to FIGS. 1-10. In some examples, a UE or a network entity may execute a set of instructions to control the functional elements of the UE or the network entity to perform the described functions.

Additionally, or alternatively, the UE or the network entity may perform aspects of the described functions using special-purpose hardware.

[0183] At block 1205, the method may include receiving a signal representing a set of multiple code blocks over a set of multiple bit channels, the set of multiple code blocks encoded using a polar code that is partitioned into a set of multiple sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme. In some examples, each bit channel of the set of multiple bit channels is associated with a respective reliability. The operations of block 1205 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of block 1205 may be performed by a bit channel component 835 as described with reference to FIG. 8.

[0184] At block 1210, the method may include de-interleaving each code block of the set of multiple code blocks using a set of multiple de-interleavers, each de-interleaver of the set of multiple de-interleavers associated with a respective code block of the set of multiple code blocks, and the set of multiple code blocks collectively including an information bit vector. The operations of block 1210 may be performed in accordance with examples as disclosed herein. In some

examples, aspects of the operations of block **1210** may be performed by a de-interleaving component **840** as described with reference to FIG. **8**.

[0185] At block **1215**, the method may include decoding the set of multiple code blocks according to the polar code to obtain the information bit vector. The operations of block **1215** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of block **1215** may be performed by a decoding component **845** as described with reference to FIG. **8**.

[0186] The following provides an overview of aspects of the present disclosure:

[0187] Aspect 1: A method for wireless communication at a wireless communication device, comprising: encoding an information bit vector into a plurality of code blocks using a polar code that is partitioned into a plurality of sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, each code block of the plurality of code blocks being encoded using a respective sub-block of the plurality of sub-blocks; interleaving respective bits of each code block of the plurality of code blocks using a respective interleaver of a plurality of interleavers; and transmitting the respective interleaved bits of each code block of the plurality of code blocks over a respective bit channel of a plurality of bit channels, each bit channel of the plurality of bit channels being associated with a respective reliability.

[0188] Aspect 2: The method of aspect 1, wherein encoding the information bit vector comprises: mapping bits of the information bit vector to respective bit locations associated with the plurality of sub-blocks, wherein the bits of each code block are interleaved in accordance with the respective bit locations.

[0189] Aspect 3: The method of any of aspects 1 through 2, wherein encoding the information bit vector comprises: allocating a respective quantity of bits of the information bit vector to each sub-block of the plurality of sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector; and determining a bit position for each bit of the respective quantity of bits allocated to each sub-block of the plurality of sub-blocks, wherein each code block of the plurality of code blocks comprises the respective quantity of bits of the information bit vector in accordance with the bit positions.

[0190] Aspect 4: The method of aspect 3, wherein allocating the quantities of bits of the information bit vector to the plurality of sub-blocks comprises: determining single-bit reliabilities of respective bit channel sub-blocks for each sub-block of the plurality of sub-blocks; selecting respective subsets of the respective bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with the single-bit reliabilities; and dividing respective sets of bits into partitions allocated to the respective subsets of the respective bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with the single-bit reliabilities.

[0191] Aspect 5: The method of aspect 4, wherein determining the bit position for each bit of the respective quantity of bits comprises: determining the bit position for each bit of the partitions in each bit channel sub-block in accordance with the single-bit reliabilities.

[0192] Aspect 6: The method of any of aspects 1 through 2, wherein allocating the respective quantity of bits of the information bit vector to each sub-block of the plurality of sub-blocks comprises: dividing respective sets of bits into partitions allocated to bit channel sub-blocks for each sub-block of the plurality of sub-blocks, wherein the respective sets of bits are divided in accordance with a mutual information transfer function, and wherein the respective quantity of bits is allocated to each sub-block of the plurality of sub-blocks in accordance with the mutual information transfer function.

[0193] Aspect 7: The method of aspect 6, wherein the respective sets of bits are divided into partitions recursively, and the recursive dividing is stopped based on the bit channel sub-blocks being associated with a threshold length, and determining the bit position for each bit of the respective quantity of bits is based on the bit channel sub-blocks being associated with the threshold length.

[0194] Aspect 8: The method of any of aspects 1 through 7, wherein each code block is associated with a respective quantity of encoded bits, the method further comprising: performing rate matching for the respective quantity of the encoded bits associated with each code block in accordance with the same block length of each sub-block.

[0195] Aspect 9: The method of aspect 8, wherein performing the rate matching for the respective quantity of the encoded bits comprises: shortening a subset of the respective quantity of the encoded bits associated with each code block, the subset comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with each code block.

[0196] Aspect 10: The method of aspect 9, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

[0197] Aspect 11: The method of aspect 8, wherein performing the rate matching for the respective quantity of the encoded bits comprises: puncturing a subset of the respective quantity of the encoded bits associated with each code block, the subset comprising a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with each code block.

[0198] Aspect 12: The method of aspect 11, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

[0199] Aspect 13: The method of aspect 8, wherein performing the rate matching for the respective quantity of the encoded bits comprises: shortening a subset of the respective quantity of the encoded bits associated with one or more code blocks from the plurality of code blocks, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with the one or more code blocks; and shortening all of the encoded bits associated with one or more other code blocks from the plurality of code blocks.

[0200] Aspect 14: The method of aspect 13, wherein the one or more other code blocks are associated with a capacity level of 1.

[0201] Aspect 15: The method of aspect 8, wherein performing the rate matching for the respective quantity of the encoded bits comprises: puncturing a subset of the respective quantity of the encoded bits associated with one or more code blocks from the plurality of code blocks, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with the one or more code blocks; and puncturing all of the encoded bits associated with one or more other code blocks from the plurality of code blocks.

[0202] Aspect 16: The method of aspect 15, wherein the one or more other code blocks are associated with a capacity level of 0.

[0203] Aspect 17: The method of any of aspects 1 through 16, wherein a total block length of the polar code is equal to N , and the same block length of each sub-block of the plurality of sub-blocks is equal to N divided by a quantity of the plurality of bit channels.

[0204] Aspect 18: The method of any of aspects 1 through 17, further comprising: mapping encoded bits to a constellation in accordance with a row-in column-out interleaving scheme, the encoded bits being mapped to the constellation prior to interleaving each code block using the respective interleaver of the plurality of interleavers.

[0205] Aspect 19: The method of any of aspects 1 through 18, wherein each interleaver of the plurality of interleavers is different in accordance with each interleaver being associated with different code blocks of the plurality of code blocks.

[0206] Aspect 20: A method for wireless communication at a wireless communication device, comprising: receiving a signal representing a plurality of code blocks over a plurality of bit channels, the plurality of code blocks encoded using a polar code that is partitioned into a plurality of sub-blocks having a same block length in accordance with a polar transformation associated with

a modulation scheme, each bit channel of the plurality of bit channels is associated with a respective reliability; de-interleaving each code block of the plurality of code blocks using a plurality of de-interleavers, each de-interleaver of the plurality of de-interleavers associated with a respective code block of the plurality of code blocks, and the plurality of code blocks collectively comprising an information bit vector; and decoding the plurality of code blocks according to the polar code to obtain the information bit vector.

[0207] Aspect 21: The method of aspect 20, wherein bits of the information bit vector are mapped to respective bit locations associated with the plurality of sub-blocks.

[0208] Aspect 22: The method of any of aspects 20 through 21, wherein a respective quantity of bits of the information bit vector is allocated to each sub-block of the plurality of sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector, and each code block of the plurality of code blocks comprises the respective quantity of bits of the information bit vector in accordance with bit positions.

[0209] Aspect 23: The method of aspect 22, wherein respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with single-bit reliabilities respectively associated with the bit channel sub-blocks, and the bit channel sub-blocks for each sub-block are selected from a plurality of bit channel sub-blocks in accordance with the single-bit reliabilities respectively associated with the bit channel sub-blocks.

[0210] Aspect 24: The method of aspect 22, wherein respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with a mutual information transfer function, and the respective quantity of the bits is allocated to each sub-block of the plurality of sub-blocks in accordance with the mutual information transfer function.

[0211] Aspect 25: The method of any of aspects 20 through 24, wherein each code block is associated with a respective quantity of encoded bits, the respective quantity of the encoded bits being rate matched in accordance with the same block length of each sub-block.

[0212] Aspect 26: The method of aspect 25, wherein a subset of the respective quantity of the encoded bits associated with each code block is shortened, the subset comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits.

[0213] Aspect 27: The method of aspect 26, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

[0214] Aspect 28: The method of aspect 25, wherein a subset of the respective quantity of the encoded bits associated with each code block is punctured, the subset comprising a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits.

[0215] Aspect 29: The method of aspect 28, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

[0216] Aspect 30: The method of aspect 25, wherein a subset of the respective quantity of the encoded bits for one or more code blocks from the plurality of code blocks is shortened, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks from the plurality of code blocks are shortened.

[0217] Aspect 31: The method of aspect 30, wherein the one or more other code blocks are associated with a capacity level of 1.

[0218] Aspect 32: The method of aspect 25, wherein a subset of the respective quantity of the encoded bits for one or more code blocks from the plurality of code blocks is punctured, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at a

beginning of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks of the plurality of code blocks are punctured.

[0219] Aspect 33: The method of aspect 32, wherein the one or more other code blocks are associated with a capacity level of 0.

[0220] Aspect 34: The method of any of aspects 20 through 33, wherein a total block length of the polar code is equal to N , and the same block length of each sub-block of the plurality of sub-blocks is equal to N divided by a quantity of the plurality of bit channels.

[0221] Aspect 35: A wireless communication device for wireless communication, comprising a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless communication device to perform a method of any of aspects 1 through 19.

[0222] Aspect 36: A wireless communication device for wireless communication, comprising at least one means for performing a method of any of aspects 1 through 19.

[0223] Aspect 37: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by one or more processors to perform a method of any of aspects 1 through 19.

[0224] Aspect 38: A wireless communication device for wireless communication, comprising a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless communication device to perform a method of any of aspects 20 through 34.

[0225] Aspect 39: A wireless communication device for wireless communication, comprising at least one means for performing a method of any of aspects 20 through 34.

[0226] Aspect 40: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by one or more processors to perform a method of any of aspects 20 through 34.

[0227] It should be noted that the methods described herein describe possible examples, and that the operations and the steps may be rearranged or otherwise modified and that other examples are possible. Further, aspects from two or more of the methods may be combined.

[0228] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0229] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0230] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (such as a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Any functions or operations described herein as being

capable of being performed by a processor may be performed by multiple processors that, individually or collectively, are capable of performing the described functions or operations.

[0231] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and examples are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0232] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media. Any functions or operations described herein as being capable of being performed by a memory may be performed by multiple memories that, individually or collectively, are capable of performing the described functions or operations.

[0233] As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0234] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. The terms “a,” “at least one,” “one or more,” “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. The term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” may refer to any or all of the one or more components. For example, a component introduced with the article “a” may be understood to mean “one or more components,” and referring to “the

component” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.” Similarly, subsequent reference to a component introduced as “one or more components” using the terms “the” or “said” may refer to any or all of the one or more components. For example, referring to “the one or more components” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.”

[0235] The term “determine” or “determining” encompasses a variety of actions and “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database or another data structure), ascertaining, among other examples. Also, “determining” can include receiving (for example, receiving information), accessing (for example, accessing data stored in memory) among other examples. Also, “determining” can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0236] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.

[0237] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0238] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. The disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A wireless communication device, comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the wireless communication device to: receive, via a plurality of bit channels, a signal representing a plurality of code blocks collectively comprising an information bit vector, the plurality of code blocks encoded in accordance with a polar code partitioned into a plurality of sub-blocks each having a same block length in accordance with a polar transformation associated with a modulation scheme, wherein each bit channel of the plurality of bit channels is associated with a respective reliability; de-interleave each code block of the plurality of code blocks using a respective de-interleaver of a plurality of de-interleavers; and decode the plurality of de-interleaved code blocks in accordance with the polar code to obtain the information bit vector.
2. The wireless communication device of claim 1, wherein bits of the information bit vector are mapped to respective bit locations associated with the plurality of sub-blocks.
3. The wireless communication device of claim 1, wherein a respective quantity of bits of the information bit vector is allocated to each sub-block of the plurality of sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector, and wherein

each code block of the plurality of code blocks comprises the respective quantity of bits of the information bit vector in accordance with bit positions.

4. The wireless communication device of claim 3, wherein respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with single-bit reliabilities respectively associated with the bit channel sub-blocks, and wherein the bit channel sub-blocks for each sub-block are selected from a plurality of bit channel sub-blocks in accordance with the single-bit reliabilities respectively associated with the bit channel sub-blocks.

5. The wireless communication device of claim 3, wherein respective sets of bits are divided into partitions allocated to bit channel sub-blocks for each sub-block of the plurality of sub-blocks in accordance with a mutual information transfer function, and wherein the respective quantity of the bits is allocated to each sub-block of the plurality of sub-blocks in accordance with the mutual information transfer function.

6. The wireless communication device of claim 1, wherein each code block of the plurality of code blocks is associated with a respective quantity of encoded bits, the respective quantity of the encoded bits being rate matched in accordance with the same block length of each sub-block.

7. The wireless communication device of claim 6, wherein a subset of the respective quantity of the encoded bits associated with each code block of the plurality of code blocks is shortened, the subset comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits.

8. The wireless communication device of claim 7, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

9. The wireless communication device of claim 6, wherein a subset of the respective quantity of the encoded bits associated with each code block of the plurality of code blocks is punctured, the subset comprising a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits.

10. The wireless communication device of claim 9, wherein the first quantity of the encoded bits is equal to a difference between a total block length of the polar code and a length of a codeword comprising the plurality of code blocks divided by a quantity of the plurality of bit channels.

11. The wireless communication device of claim 6, wherein a subset of the respective quantity of the encoded bits for one or more code blocks from the plurality of code blocks is shortened, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at an end of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks from the plurality of code blocks are shortened.

12. The wireless communication device of claim 11, wherein the one or more other code blocks are associated with a capacity level of **1**.

13. The wireless communication device of claim 6, wherein a subset of the respective quantity of the encoded bits for one or more code blocks from the plurality of code blocks is punctured, the subset of the respective quantity of the encoded bits comprising a first quantity of the encoded bits at a beginning of the respective quantity of the encoded bits associated with the one or more code blocks, and all of the encoded bits associated with one or more other code blocks of the plurality of code blocks are punctured.

14. The wireless communication device of claim 13, wherein the one or more other code blocks are associated with a capacity level of **0**.

15. The wireless communication device of claim 1, wherein a total block length of the polar code is equal to N , and wherein the same block length of each sub-block of the plurality of sub-blocks is equal to N divided by a quantity of the plurality of bit channels.

16. A method for wireless communication by a wireless communication device, comprising:

receiving, via a plurality of bit channels, a signal representing a plurality of code blocks collectively comprising an information bit vector, the plurality of code blocks encoded in accordance with a polar code partitioned into a plurality of sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, wherein each bit channel of the plurality of bit channels is associated with a respective reliability; de-interleaving each code block of the plurality of code blocks using a respective de-interleaver of a plurality of de-interleavers; and decoding the plurality of de-interleaved code blocks in accordance with the polar code to obtain the information bit vector.

17. The method of claim 16, wherein bits of the information bit vector are mapped to respective bit locations associated with the plurality of sub-blocks.

18. The method of claim 16, wherein a respective quantity of bits of the information bit vector is allocated to each sub-block of the plurality of sub-blocks in accordance with reliabilities respectively associated with the bits of the information bit vector, and wherein each code block of the plurality of code blocks comprises the respective quantity of bits of the information bit vector in accordance with bit positions.

19. The method of claim 16, wherein each code block of the plurality of code blocks is associated with a respective quantity of encoded bits, the respective quantity of the encoded bits being rate matched in accordance with the same block length of each sub-block.

20. A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by one or more processors to: receive, via a plurality of bit channels, a signal representing a plurality of code blocks collectively comprising an information bit vector, the plurality of code blocks encoded in accordance with a polar code partitioned into a plurality of sub-blocks having a same block length in accordance with a polar transformation associated with a modulation scheme, wherein each bit channel of the plurality of bit channels is associated with a respective reliability; de-interleave each code block of the plurality of code blocks using a respective de-interleaver of a plurality of de-interleavers; and decode the plurality of de-interleaved code blocks in accordance with the polar code to obtain the information bit vector.
